

Urban Economics

Urban Sprawl and Land-Use Controls

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Outline for Today

1.



Next week

Readings



Introduction





Urban sprawl has been drawing strong feelings

Critics say:

- Increases traffic congestion and pollution 🚗
- Consumes too much land 🌲
- Less incentives to develop downtown areas 🏢
- Low density sprawl reduces social interaction 👤👤
- Dependence on automobiles increases obesity rates

The data shows:

- Spatial growth was really high---from 1.3% to 1.9% of land area got developed from 1976 to 1992 (Burchfield et al., 2006) 📈
- Other sources show that the rate is even higher

As a result, many cities have implemented land-use controls to limit sprawl



Growth management policies to control urban sprawl

At least 12 states have growth management policies

New Jersey committed in 1998 to spend \$1 billion to purchase vacant land to limit



Do the critics have a point?

Do cities take up too much space?

Should they implement land-use controls to limit sprawl?

The Empirical Evidence



This table shows the results from two papers

Elasticities of urban area land with respect to various variables.

	Brueckner and Fansler (1983)	McGrath (2005)
Population (L)	1.10	0.76
Commuting cost (t)	0	-0.28
Income (y)	1.50	0.33
Agricultural rent (r_A)	-0.23	-0.10

The results tells us if X increases by 1 percent, Y increases by how much percent?

Both studies show that higher population leads to more land use

- 1 percent increase in population leads to 1.1 and 0.76 percent increase in land use

Income effects are large: 1 percent increase in income leads to 1.5 and 0.3 percent increase in land use

Higher agricultural rent leads to less land use, mixed evidence on the effect of commuting costs

It is easy to see why urban sprawl happens in the US

So, if we are growing in population and getting richer, is it bad that we are sprawling out?

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So, if we are growing in population and getting richer, is it bad that we are sprawling out?

Well, it depends

If this sprawl is being driven by consumer preferences, then it is **not a problem**

However, if sprawl is being driven by market failures, then it is a **problem**

Market failures related present in economic process that determines size of cities

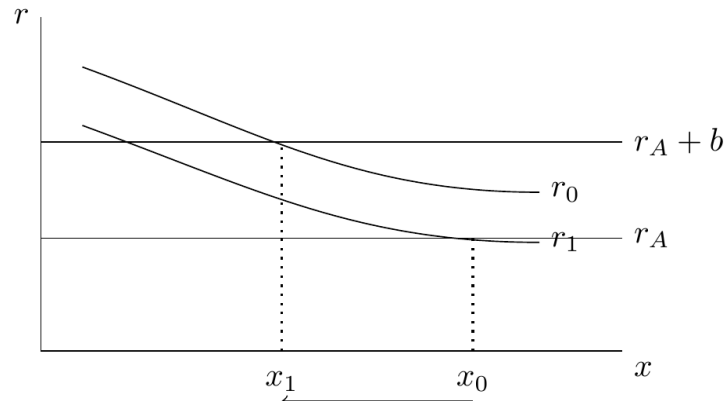
1. The amenity value of open space

- Assume each acre of open space provides b dollars above and beyond its agricultural value
- This benefit is not captured when land is converted from agricultural to urban use
- A landowner can either rent the land to a developer r rent it out to a farmer r_a
- If $r > r_a$, the land will be developed
- Society would ideally want the landowner to take into account the amenity value b
- So society would want the land developed only if $r + b > r_a \Rightarrow$ city boundary rent should be $r = r_a - b$

Socially optimal effect agricultural rent

1. The amenity value of open space

- So a optimal city size would emerge if the rent is $r = r_a - b$, which includes the amenity value of open space



- How do we get from x_0 to x_1 ?
 - By implementing land-use controls that limit development
 - Imposing a development tax

This model assumes that people value open space and the government knows the value

Is this realistic?

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Is this realistic?

It is not hard to imagine that the person designing anti-sprawl policies might reach a value that is larger than that of the average person

This could lead to reducing dwelling sizes and make living in the city more costly (people do not value the open space that is saved)

So land-use controls could lead to a loss in welfare if the value of open space is overestimated



Market failures related present in economic process that determines size of cities

2. The market failure related to traffic congestion



Market failures related present in economic process that determines size of cities

2. The market failure related to traffic congestion

Why congestion is a market failure

- Each additional car on a congested road **slows down all other drivers**
- Drivers bear only their **private commuting cost**, not the **external delay** they cause
- Result: The **private cost < social cost**, leading to **too much driving** and **excessively large cities**

The Congestion Externality

- The *externality* arises because a single extra car increases travel time for all others
- Each driver's additional time loss is small, but **summed across all drivers**, the total cost is **non-negligible**
- No individual driver has the incentive to account for this social cost
- Thus:
 - Commuting decisions are distorted
 - City size is **too large** in the free-market equilibrium

The Effect on Urban Spatial Structure

Without congestion tolls:

- Commuting is **too cheap**
- Residents are willing to live farther away
- The equilibrium city radius would be **larger than optimal**

With congestion tolls:

- Higher commuting cost → shorter optimal commutes
- Population locates **closer to the center**
- The city becomes **more compact**

Congestion Tolls as a Policy Tool

What the toll should equal:

- A congestion toll must equal the **external cost** a driver imposes on others
- When this toll is charged:
 - Commuters internalize the full social cost
 - Traffic volumes fall
 - The **city shrinks to the efficient size**

Key insight:

Tolls make the market mimic the social optimum.

Evidence from Real-World Congestion Pricing

Cities implementing congestion pricing:

- **Singapore** (one of the earliest and most comprehensive systems)
- **London** (central-city cordon charge)
- **Stockholm** (peak-hour bridge tolls)
- **New York City**

What studies find:

- After toll adoption, cities become **more compact**
- Residents choose **shorter commuting distances**
- Urban models correctly predict **declines in city size** when tolls are introduced

From the News

The New York Times

New York Today

The Streets Are Safer This Year for Drivers, Cyclists and Pedestrians

There have been 87 traffic deaths in the city so far this year, the lowest number since 2018. Officials credit the Vision Zero program and congestion pricing.

Local News

Congestion pricing reducing traffic in NYC, increasing revenue after first 6 months, officials say

By Jesse Zanger

Welfare Effects of Correcting Congestion Externalities

- By increasing the cost of commuting, tolls:
 - Reduce excessive driving
 - Align **private** and **social** commuting costs
 - Lead to **higher welfare** than the no-toll equilibrium
- Since congestion makes free-market cities **too large**,
charging congestion tolls moves the city toward the socially optimal spatial size.

Land-Use Controls: Policies





Using Land-Use Controls to Attack Sprawl

Two main anti-sprawl policies discussed:

1. **Development tax**

- Price-based instrument
- Internalizes the open-space externality (amenity benefits)

2. **Congestion toll**

- Corrects the congestion externality
- Shrinks the city by raising effective commuting costs

Urban Growth Boundary (UGB)

What is a UGB?

- A **quantity-based** anti-sprawl instrument
- Government sets a maximum city radius $\bar{x}_1$
- Development outside this boundary is prohibited

Key point:

A UGB can replicate the **same effect** as a development tax *if set at the optimal radius $\bar{x}_1$.

- In the model, restricting development at $\bar{x}_1$ produces the same land-rent curve shift as imposing a per-acre development tax b .

Why UGBs Can Be Problematic

- A UGB limits **land area**, not commuting decisions
- Thus, it corrects the symptom (too much land consumption), not the cause (externality)
- When the externality is **traffic congestion**, a UGB **cannot substitute** for a congestion toll

Important distinction:

- A development tax fixes the underlying externality
- A UGB **mimics the outcome** only in the open-space amenity case
- But it **fails** in the congestion case because it does not change commuting incentives

UGBs and Consumer Welfare

When UGBs can reduce welfare:

- If average consumers do **not** value preserved open space highly
- Limiting housing supply increases:
 - **Land rents**
 - **Housing prices**
 - **Density**
 - **Living costs**

Policy implication:

UGBs should be used with caution because the welfare costs may exceed the benefits.

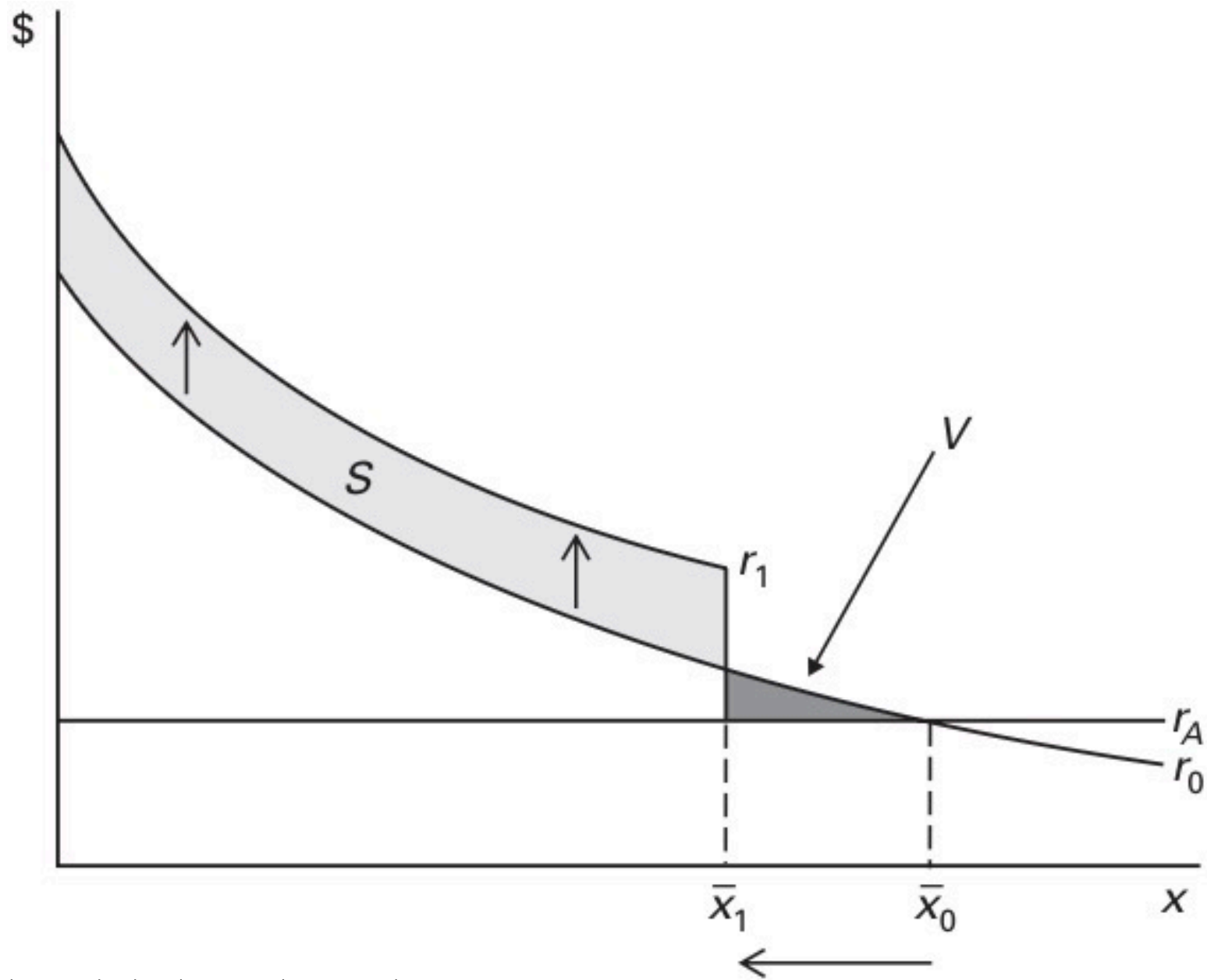
A Motive for UGBs: Landowner Gain

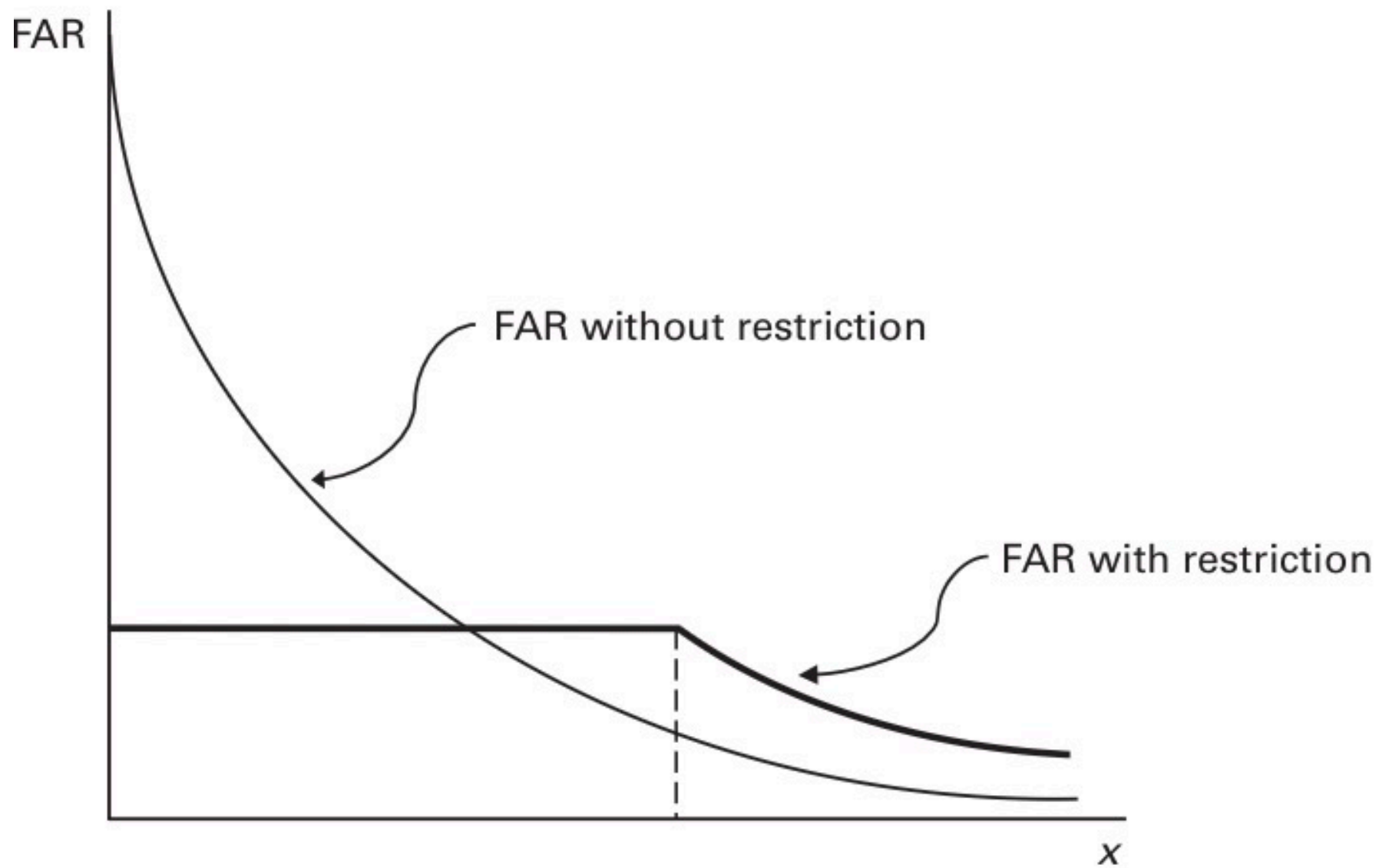
Political economy explanation

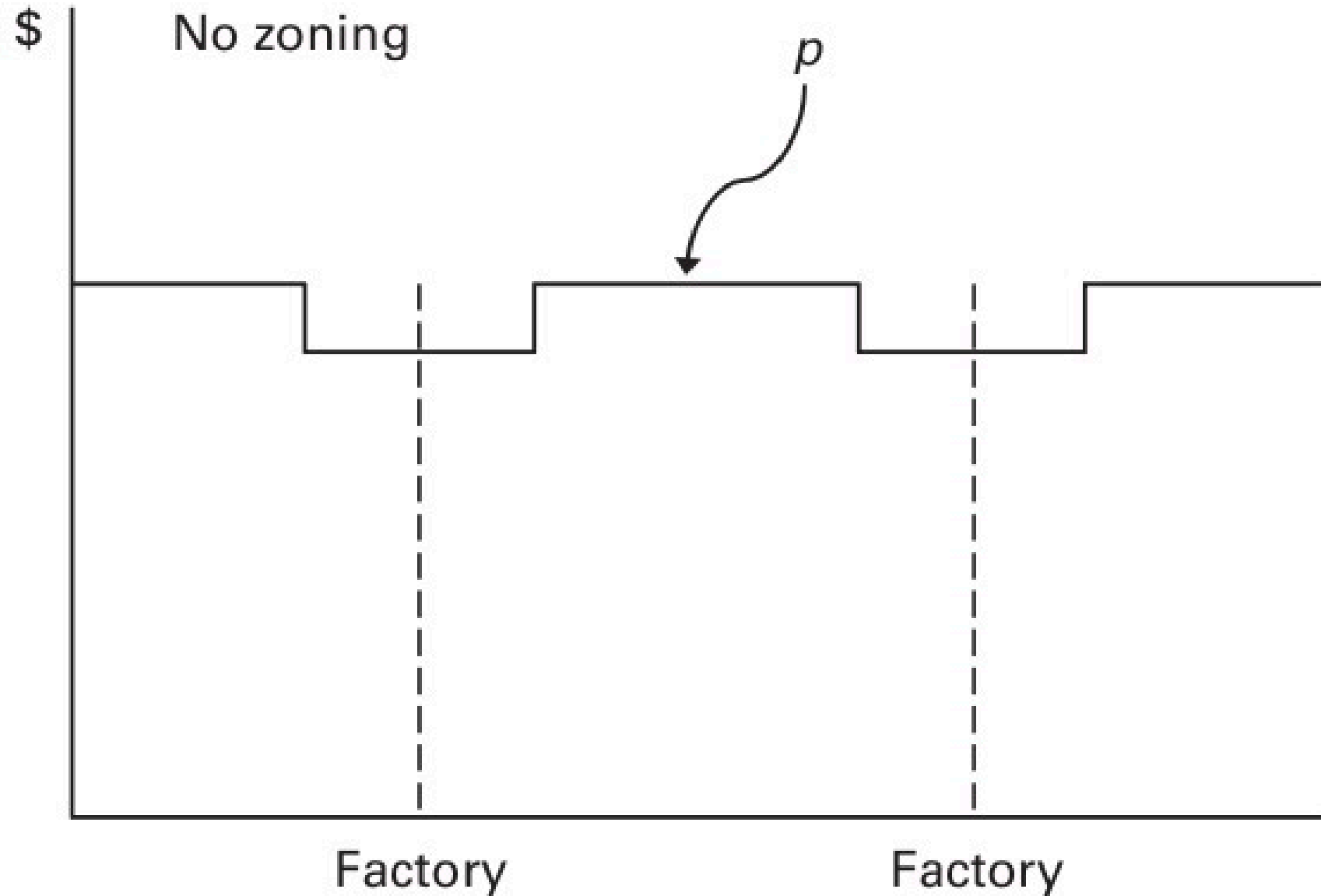
- Landowners **gain** when UGB raises land rents inside boundary
 - Restricting city size shifts the rent curve up
 - Generates rent gains represented by area **S**
 - These gains accrue to landowners
- As long as the rent gain **S** > **V** (the loss outside the boundary), landowners push for tighter UGBs

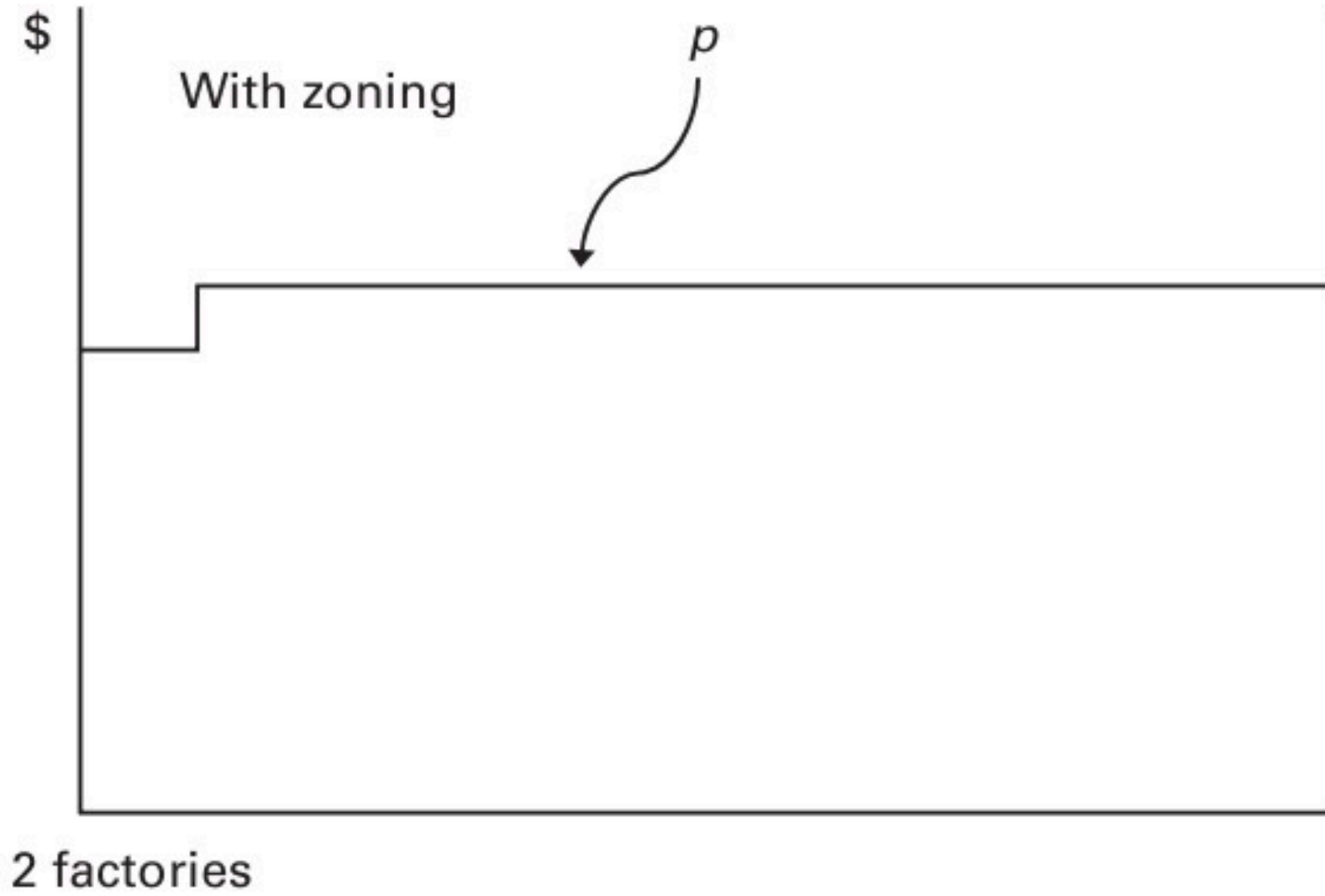
Result:

UGBs may be adopted **not** for efficiency reasons, but because landowners benefit financially.









Additional Motives for UGBs

- Some motivations are **not** about market failures:
 - Desire to boost housing capital gains
 - Local political pressure to preserve aesthetics or neighborhood character
 - Resistance to density and “undesirable” growth

Key point:

Policies may be driven by politics, not economics.



Building-Height and Density Restrictions

Height limits:

- Many cities impose max building heights or Floor Area Ratio (FAR) limits
- Examples:
 - Washington, DC
 - Paris
 - Mumbai and Bangalore

Effect in the urban model:

- Lower allowed FAR leads to:
 - Smaller buildings in the city center
 - Housing becomes more expensive
 - Residents consume smaller dwellings
 - City expands spatially → **more sprawl**

Takeaway:

Building-height limits **increase** sprawl by reducing density where people most want to live.

FAR Restrictions: Model Implications

- When FAR is restricted:
 - Housing prices p increase
 - Land rents r increase
 - The city's boundary $\bar{x}$ expands
 - Dwellings get smaller
 - Outer areas see **taller buildings** (nonbinding FAR)

Irony:

Policies intended to reduce density at the core **increase overall sprawl**.

Minimum Lot Size Rules

- Common in suburban communities
- Example: min lot size of $\frac{1}{2}$ acre for a detached house
- These effectively **limit population density**
- Result:
 - Higher land consumption per household
 - Larger city footprint
 - Additional sprawl

Zoning Laws

Most pervasive land-use control in U.S.

- Separate land uses into zones:
 - Residential
 - Commercial
 - Industrial
 - High/low density classifications

Purpose:

Prevent negative externalities (noise, pollution, traffic)

Example:

- Factories generate noise/dirt → lower nearby housing prices
- Zoning places factories at city's edge to minimize externalities

Zoning and Land Rents

- Without zoning:
 - Factories locating near residential areas lower p nearby
 - Land rents fall across large neighborhoods
- With zoning:
 - Factories are consolidated in one area
 - Negative externality footprint reduced by ~75%
 - Landowner income loss decreases proportionally

Takeaway:

Zoning can improve welfare by **reducing negative externalities**.

Is Zoning Necessary?

- Some cities impose **no zoning at all** (e.g., Houston)
- Houston's development patterns don't look dramatically different from zoned cities
- Argument:
 - Externalities sometimes manage themselves
 - Market forces may naturally separate incompatible land uses

But:

- Houston's growth was unusual (large-scale single-ownership developments)
- These internal arrangements acted *like* zoning substitutes



Empirical Evidence on Land-Use Controls

Findings from the literature:

- UGBs
- Height restrictions
- Density restrictions
- Impact fees

➔ **All tend to raise housing prices.**

Why?

- They restrict supply
- They increase scarcity of developable land
- They reduce the elasticity of housing production